Precision and recall are therefore based on an understanding and measure of relevance.

It is possible to give an alternate definition of precision and recall by introducing the number of *false positives* Fp, i.e. points belonging to the `false` class for which `true` was incorrectly predicted) and *false negatives* Fn.

$$precision = \frac{T_P}{T_P + F_P}$$

$$recall = \frac{T_P}{T_P + F_N}$$

Intuitively, while accuracy only measures how well we do on predictions, precision/recall metrics weight our successes with our errors for the twin category. As a matter of fact, precision and recall are not independent and one can be defined in terms of the other, so it’s not possible to improve both indefinitely.

For *Machine Learning* classifiers, every model can be associated with a precision/recall curve and its parameters can be tuned during training so that we can trade off the two characteristics.

So, for binary categorization, a perfect (100%) precision means that every point for which we output `true` actually belonged to the true category. Similarly for recall, when it's 100% we never have a false negative.

As an exercise, try to compute precision and recall for the examples in the previous sections, and see how they give us